

Visual RAG

An Autonomous Visual Search Agent for Video Retrieval

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Executive Summary

Video is widely recorded but remains hard to search at the moment level. The questions people ask of footage are causal and temporal — *why* did the boy carry the present to the sofa — while the metadata and speech transcripts video search relies on are neither. This dissertation asks how far an agentic retrieval-augmented generation system can close that gap with off-the-shelf components on consumer hardware, treating every design choice as something to measure rather than assert.

The system indexes visual and transcript evidence as overlapping 8-second segments, rewrites a question into the scene-caption register a contrastive encoder can match, searches and re-ranks candidate moments, and answers through a bounded LangGraph agent that cites the seconds of video supporting each claim. Retrieval is scored on 3,358 temporally grounded NExT-QA/NExT-GQA questions over 567 videos under a deliberately strict criterion — correct video *and* tIoU ≥ 0.5 — which makes absolute recall small by construction and puts the weight on controlled comparisons.

- A reproducible, temporally grounded video RAG pipeline and evaluation harness, runnable end to end on one 6 GB laptop GPU (Chapters 3–4).
- A component-wise ablation over all 3,358 questions: a SigLIP index with query decomposition nearly doubles top-1 moment recall (0.026 to 0.048); decomposition buys recall at depth, visual re-ranking top-rank precision (§5.3).
- Two negative results with one diagnosis — late fusion never beats visual-only at any weight, and a text cross-encoder re-ranker degrades ranking, both traced to sparse, multilingual, conversational speech — together with a cross-domain replication on speech-dense video that reverses both exactly as the diagnosis predicts, making the modality ordering domain-conditional and measurable (§5.3.1, §5.3.3, §5.5).
- A two-stage equivalence finding: re-ranking a cheap index’s top-30 with a stronger backbone matches indexing the whole corpus with it, decoupling quality from indexing cost (§5.3.4).
- A calibrated account of agentic answering: the graph agent lifts five-choice accuracy 0.447 to 0.547 and keyframe evidence lifts temporal-next accuracy 0.341 to 0.636, while prior-only baselines (0.560) leave the multimodal stack the only configuration that beats the bare model (§5.4).

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I would like to thank my supervisor, Dr Junxiao Shen, for proposing this project and for guidance that consistently sharpened it. My sincere thanks also go to my friend Zhuolun Li, whose generous help throughout the project made a real difference, and to my classmates for their support and discussion.

Statement on the inclusion of published and collaborative work

My dissertation does not include work that has been published, submitted or prepared for publication with me as an author. My dissertation does not include non-published collaborative work that contains data or contributions from others.

Author's Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

A handwritten signature in black ink, reading "Zhenyu Zhang". The script is cursive and fluid, with the first name "Zhenyu" and the last name "Zhang" clearly distinguishable.

Zhenyu Zhang

18 July 2026

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1 Introduction

1.1 Motivation

Video has become the dominant medium in which everyday life, education, and work are recorded, yet it remains the least searchable. Text search operates at the granularity of the answer: a query returns the sentence or passage that contains it. Video search, in practice, still operates at the granularity of the *container*: a title, a description, at best a manually chaptered timeline. The question a user actually has — *why did the boy carry the present to the sofa? what did the dog do after it reached the cushion?* — is a question about a particular few seconds of footage, phrased in causal and temporal language that neither keyword matching over metadata nor speech-transcript search can resolve. The gap is felt anywhere video accumulates faster than it can be watched: personal libraries, lecture archives, meeting recordings, broadcast footage, and the industrial video-intelligence platforms that motivated this project’s supervision.

Two research threads have recently made this gap addressable. Contrastive vision-language models (CLIP and its successors) embed images and natural language into a shared space, making *what is on screen* directly searchable by description. And retrieval-augmented generation (RAG) shows how a large language model can be grounded in retrieved evidence to answer questions it could not answer alone — with the retrieval unit, in the video case, naturally being a *time-stamped segment* rather than a document. Between these threads sits an under-explored composition: an *agentic* system that decomposes a causal or temporal question, searches a multi-modal index of video segments, reasons over what it retrieves — including reasoning *along the timeline* — and synthesises an answer whose every claim cites the seconds of video that support it.

This dissertation designs, builds, and evaluates such a system end-to-end, under a deliberately modest compute budget (a single consumer laptop GPU plus commodity LLM APIs), and subjects each of its design decisions to quantitative test on a benchmark

with ground-truth temporal annotations.

1.2 Research questions

The project plan fixed three research questions, which this dissertation answers in Chapter 5:

- **RQ1** — How effectively can a Visual RAG pipeline retrieve relevant video segments from natural-language queries compared to text-only retrieval baselines?
- **RQ2** — What is the optimal embedding strategy for multi-modal video representation (visual + audio + text)?
- **RQ3** — How can an agentic framework autonomously decompose complex queries into sub-retrieval tasks and synthesise coherent answers?

A scoping note on RQ2: in the system evaluated here the audio channel enters as *transcribed speech* (Whisper ASR), so “text” throughout this dissertation means the spoken content of the video. Optical character recognition of on-screen text is implemented in the ingestion pipeline but remained disabled for every experiment and is out of scope; the modality comparisons of Chapter 5 are therefore between visual embeddings and speech-transcript embeddings.

Temporal precision is treated throughout as a first-class requirement rather than an afterthought: retrieval is scored not on finding the right *video* but on finding the right *moment* (temporal intersection-over-union against ground-truth intervals), and generated answers must carry timestamp citations. One consequence deserves stating before any numbers are read. A retrieval *hit* requires the correct video **and** tight temporal overlap ($\text{tIoU} \geq 0.5$) between a fixed 8-second window and a freely placed ground-truth interval — found among all 5,725 segments of the corpus. This is a far stricter criterion than the video-level retrieval measures common in the literature, and it makes absolute scores small *by construction*: corpus-scope top-1 recall in the low single digits of a percent is the expected scale on this metric, not a symptom of a broken system. Results throughout should accordingly be read as controlled comparisons on a deliberately hard metric, and Chapter 5 restates this where the first numbers appear.

1.3 Contributions

The dissertation makes six contributions, each developed in the chapter noted and empirically grounded in Chapter 5:

1. **An end-to-end, openly reproducible video RAG pipeline with temporal grounding** (Chapter 3): ingestion (keyframes, Whisper speech recognition), overlapping temporal segmentation, dual-modality indexing in a joint vision-language space, staged retrieval, and an answering agent with per-claim timestamp citations — evaluated on 3,358 temporally annotated questions over 567 videos (NExT-QA/NExT-GQA), and runnable end-to-end on one consumer GPU plus approximately US\$5 of API usage for the retrieval and text-only answering programme (approximately US\$30 including the multimodal best-configuration run of §5.4.4).
2. **A systematic ablation of the multi-modal embedding strategy** (§5.3), including two documented *negative results* with causal analyses: weighted late fusion of visual and transcript embeddings never outperforms visual-only retrieval at any weight on this domain, and a text cross-encoder re-ranker strictly degrades ranking quality. Both trace to the same cause — sparse, multilingual, conversational speech is a poor retrieval key for everyday video — and both redirect effort toward the visual channel. A 40-video speech-dense replication (§5.5, YouCook2) then confirms the causal reading: both negative results reverse sign exactly as predicted, which establishes the modality ordering as domain-conditional and measurable rather than universal.
3. **A two-stage retrieval equivalence finding** (§5.3.4): re-ranking a cheap index’s top-30 with a stronger visual backbone matches indexing the entire corpus with that backbone to within measurement noise, while exchanging the index for SigLIP nearly doubles top-1 moment retrieval — together giving a concrete, measured recipe for scaling quality and cost independently.
4. **An agentic answering layer with temporal tools and bounded self-reflection** (Chapter 3, §5.4): a LangGraph state machine that anchors events by search, walks the timeline around them, and audits its own draft citations

against gathered evidence, improving five-way QA accuracy from .447 to .547 over a single-loop agent. A controlled evidence-channel experiment, holding tools, prompts and retrieval fixed, then shows that supplying keyframes to the answering model nearly doubles accuracy on *what-happened-next* questions (.341 $\rightarrow$.636). Modality therefore matters independently at both ends of the pipeline.

5. **A prior-only calibration of those gains** (§5.4.3), which reframes what the agent machinery buys. Given no evidence at all, both LLMs already answer at .560, so the .200 random floor is the wrong bar. Against the right bar the best text-only stack merely matches its own prior and falls .227 below it on temporal-next questions, while the multimodal stack exceeds its prior by +.167 — making visual evidence the only channel measured that beats the bare model, and delivered-but-irrelevant evidence measurably worse than none.
6. **Reproducible research artefacts:** the complete codebase, evaluation harness, per-experiment result files, provenance log mapping every reported number to the command that produced it, generated figures, and an interactive demonstration interface whose controls map one-to-one onto the ablation dimensions.

1.4 Dissertation structure

Chapter 2 reviews the background: contrastive vision-language pretraining, video moment retrieval and its benchmarks, retrieval-augmented generation, and agentic LLM frameworks. Chapter 3 presents the system design with the rationale for each decision. Chapter 4 records the implementation: the concrete stack and the engineering required to run the full pipeline within the stated budget. Chapter 5 — the core of the dissertation — evaluates the system against the three research questions. Chapter 6 discusses what the results mean, their limitations, and threats to validity. Chapter 7 concludes and outlines future work.

2 Background and Related Work

This chapter reviews the four research threads the system composes — contrastive vision-language pretraining (§2.1), video moment retrieval and its benchmarks (§2.2), retrieval-augmented generation (§2.3), and agentic LLM frameworks (§2.4) — then surveys the emerging work that, like this project, combines them into LLM-driven video understanding systems (§2.5), and states the gap this dissertation addresses (§2.6). Citation keys refer to `references.bib`.

2.1 Contrastive vision-language pretraining

CLIP [1] established that a dual-encoder trained with a contrastive objective on web-scale image-text pairs yields a joint embedding space in which arbitrary natural-language descriptions can be matched against images zero-shot. Two properties of this family underpin the present system. First, the *dual-encoder* architecture separates image and text encoding, so a corpus can be embedded once offline and queried later with only the (cheap) text tower — the architectural premise of Chapter 3’s offline/online split. Second, because training pairs are alt-text captions, the encoders are strongest at matching *literal scene descriptions*; abstract, causal, or temporal language falls outside the training distribution, a mismatch this project addresses at query time (§3.4) rather than by fine-tuning.

The open-source replication effort around `open_clip` demonstrated reproducible scaling laws for this family and released the LAION-trained checkpoints used here [2, 3]. SigLIP [4] replaced CLIP’s softmax-based InfoNCE objective with a pairwise sigmoid loss, removing the global normalisation over the batch and yielding markedly stronger retrieval performance at comparable scale. The backbone ablation of §5.3.4 — in which SigLIP SO400M nearly doubles top-1 moment retrieval over CLIP ViT-B — provides an independent, temporally grounded replication of that advantage on everyday video.

2.2 Video moment retrieval, video QA, and temporal grounding

Text-to-video retrieval initially operated at whole-video granularity, with benchmarks such as MSR-VTT [5]; CLIP4Clip [6] showed that frame-level CLIP features, pooled, transfer to this task with minimal video-specific machinery. That is the same frames-as-proxy assumption this system makes at segment level. *Moment* retrieval sharpened the task to localising an interval inside a video: Gao et al. [7] introduced the sliding-window formulation and Charades-STA, dense event captions came with ActivityNet Captions [8], and Moment-DETR [9] reframed localisation as end-to-end set prediction. The present system inherits the sliding-window formulation, its 8-second overlapping segments being windows scored by embedding similarity. It performs *corpus-level* moment retrieval, however: the query does not name its video, so finding it is part of the task. That setting is an established sub-task in its own right, video corpus moment retrieval, introduced with TVR [10] and normally studied with in-domain supervision and subtitle-aligned training data. The difference here is not the task but the means: the same retrieval is served zero-shot by composed off-the-shelf encoders, and scored by tIoU against grounding labels rather than by the benchmark’s own retrieval metric.

On the question-answering side, NExT-QA [11] moved video QA beyond descriptive questions to causal and temporal ones (*why* and *what-happened-next*) over everyday videos. Its extension NExT-GQA [12] added 10.5K temporal grounding labels and reported a sobering finding: state-of-the-art video-language models answer well while attending to the *wrong moments*, i.e. their answers are weakly grounded. This finding motivates both the benchmark choice and the evaluation design of this dissertation. NExT-GQA’s interval labels are what allow retrieval to be scored at moment granularity (tIoU) rather than video granularity, and answer grounding, meaning every claim citing its seconds of video, is treated as a first-class output requirement rather than a diagnostic. SeViLA [13] addressed grounding by chaining a localiser and an answerer within one trained model; the present system reaches a similar localise-then-answer decomposition, but with zero-shot components orchestrated by an agent instead of end-to-end training.

2.3 Retrieval-augmented generation

RAG couples a parametric language model with a non-parametric retrieval index [14, 15], classically over dense passage embeddings [16]. Three RAG sub-problems recur in this dissertation in video form.

Query–document mismatch. When queries and indexed content live in different registers, rewriting the query helps: HyDE [17] generates a hypothetical document and searches with its embedding; least-to-most prompting [18] decomposes complex questions into simpler sub-problems. The W5 decomposer (§3.4) transfers this idea across modalities: the LLM rewrites a causal question into the *scene-caption register that a contrastive visual encoder can match*, which is HyDE’s manoeuvre with “document” replaced by “caption”. Multiple rewrites are combined with reciprocal rank fusion [19], whose rank-based, score-free formulation is what makes fusing differently phrased sub-queries safe.

Precision at the top of the ranking. Two-stage retrieve-then-re-rank pipelines with a cross-encoder second stage are standard in text retrieval [20], with the BGE family [21] among the strongest open re-rankers. Chapter 5 shows the transfer to video is *not* automatic: the text cross-encoder degrades ranking on everyday video because transcripts are weak relevance evidence (§5.3.3), and the effective second stage is instead a stronger *visual* encoder — same architecture-level pattern, different modality.

Self-critique. Self-RAG [22] trains a model to critique its own retrieval and generation with reflection tokens. The graph agent’s reflect node (§3.5) implements the same principle — audit the draft against the retrieved evidence before answering — as a zero-shot prompted pass with an explicit, bounded control-flow position, rather than a trained capability.

2.4 Agentic LLM frameworks

Chain-of-thought prompting [23] established that intermediate reasoning improves LLM task performance; ReAct [24] interleaved that reasoning with *actions* against external tools, defining the observe–think–act loop that underlies most current agents, including both agents in this project. Toolformer [25] demonstrated that tool invocation can be learned self-supervised, foreshadowing the native function-calling interfaces that

2 Background and Related Work

commercial LLM APIs now expose and on which the W4 agent is built directly. Reflexion [26] added verbal self-feedback across episodes, improving agents without weight updates — the within-episode analogue of which is the reflect node above. This project implements its multi-step agent on LangGraph (a state-machine orchestration library from the LangChain project), chosen because it makes the control flow — nodes, bounded loops, termination conditions — explicit and inspectable (§3.5), in contrast to prompt-encoded loops whose behaviour is implicit in model outputs.

2.5 LLM-driven video understanding and video RAG

A rapidly growing line of work applies LLM orchestration to video. LLoVi [27] shows a deliberately simple recipe — densely caption short clips, then let an LLM aggregate the captions — is strong on long-video QA including NExT-QA. Two contemporaneous systems named VideoAgent make the LLM an active controller: Wang et al. [28] iteratively *search* for informative frames, answering NExT-QA with ~ 8 frames on average, while Fan et al. [29] equip the LLM with a structured memory over events and object tracks, queried through tools. VideoRAG [30] scales the RAG framing to extremely long multi-video corpora with a graph-organised index. The present system sits in this family but differs in emphasis in three ways. First, *temporal grounding is the deliverable*: retrieval is evaluated by tIoU against NExT-GQA intervals and answers must cite timestamps, whereas the systems above report QA accuracy alone. Second, it operates at *corpus* scope — the question does not name its video, and finding it is part of the task. Third, its contribution is *measurement*: a component-wise ablation programme (modality, fusion, decomposition, re-ranking, backbone, evidence channel) with documented negative results, rather than a new model or a leaderboard entry.

2.6 Summary of the gap

Each ingredient of this project is established: contrastive encoders for zero-shot visual matching, sliding-window moment retrieval, RAG’s retrieve-rewrite-rerank toolkit, ReAct-style tool agents, and LLM video orchestration. What the literature lacks — and what NExT-GQA’s grounding finding [12] shows is missing in practice — is an account of how these ingredients compose into a *temporally grounded*, corpus-scope video RAG

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system, with the contribution of each component measured under one protocol, including where standard components fail to transfer. Providing that account, on reproducible consumer-scale infrastructure, is the role of Chapters 3–6.

3 System Design

This chapter describes the design of the Visual RAG system and, for each component, the reasoning behind its design choices. The architecture was shaped by one deliberate constraint and one methodological commitment. The constraint is that the online system must run on consumer hardware (a 6 GB laptop GPU): all heavy computation is pushed into a one-time offline indexing stage, and the per-query path touches only a text encoder, an approximate-nearest-neighbour index, and an LLM API. The commitment is that every retrieval component should be *independently measurable*: modalities are indexed separately, pipeline stages are toggleable, and each addition of Chapters 4–5 (decomposition, re-ranking, temporal tools, reflection) can be switched on and off without touching the rest. This is what makes the ablation programme of Chapter 5 possible, and several design choices below are justified primarily by it.

3.1 Architecture overview

Figure 3.1 shows the system’s two lanes. The **offline lane** ingests a video corpus and produces a portable index: keyframes and timestamped transcripts are extracted per video, fused into overlapping temporal segments, embedded into a joint vision-language space, and stored in a vector database. The **online lane** serves a natural-language query: the query is (optionally) decomposed into retrieval-friendly sub-queries, encoded, searched against the index, (optionally) re-ranked by a stronger visual model, and the resulting segments — timestamps, transcripts, and keyframes — are handed as evidence to an answering agent that produces a grounded response with per-claim timestamp citations.

The offline/online split is more than an engineering convenience. It makes the expensive artefacts *reusable and portable* — the index of 567 videos is a few hundred megabytes that can be computed once on any GPU and copied anywhere — and it makes the research

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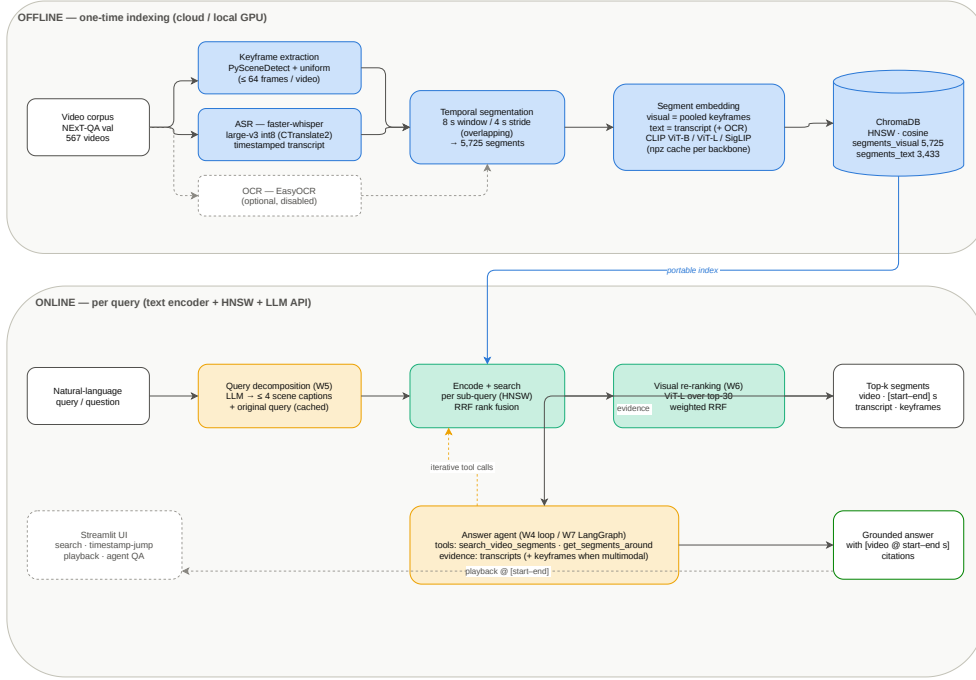


Figure 3.1: Visual RAG offline indexing and online retrieval pipeline.

loop fast: every experiment in Chapter 5 re-uses the same frozen ingestion outputs, so a full retrieval evaluation over 3,358 questions runs in minutes on a laptop. The split also mirrors production video-search architectures, where indexing throughput and query latency are separate budgets.

3.2 Ingestion and temporal segmentation

Keyframes. Each video is decoded once and sampled by two complementary strategies: shot-boundary detection (PySceneDetect’s adaptive detector) contributes a representative frame per scene change, and uniform sampling at 0.5 fps guarantees coverage of long static shots that trigger no boundary. Timestamps are de-duplicated with a minimum 1 s gap and capped at 64 frames per video (an even subsample if exceeded); frames are stored as JPEGs resized to 384 px on the long side. The dual strategy reflects the

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two failure modes of either alone: scene detection under-samples static footage, uniform sampling misses brief transitions.

Speech. Audio is transcribed by Whisper [31] large-v3 in int8 quantisation via CTranslate2, yielding sentence-level chunks with start/end timestamps. Two robustness details matter in practice. First, videos without an audio stream are detected up front and skipped rather than crashing the batch — a real failure mode discovered with silent test clips. Second, language is auto-detected per video and *not* forced to English: NExT-QA’s home videos are multilingual, and transcribing them faithfully (rather than mistranscribing them as English) preserves the option of translate-then-embed pipelines later, at the cost of the encoder mismatch analysed in §5.2. On-screen-text OCR is implemented but disabled by default: benchmark footage carries almost no legible text, and the modality can be enabled by configuration where it earns its cost.

Temporal segmentation. The atomic unit of retrieval is the *segment*: an 8-second window advanced with a 4-second stride, carrying every keyframe whose timestamp falls inside it, the concatenated transcript chunks that overlap it, and its exact [start, end] interval. Windows with neither a frame nor text are dropped; the 567-video corpus yields 5,725 segments. Three considerations fix these numbers. The window must be short enough that “retrieving the segment” constitutes a temporally precise answer — 8 s against a mean video length of 44 s gives roughly a five-fold refinement over whole-video retrieval, and is the right order of magnitude for NExT-GQA’s ground-truth moments. The 50% overlap ensures that an event near a window boundary appears *whole* in at least one window, at the cost of near-duplicate retrieval results (adjacent overlapping windows frequently co-occur in top ranks, a benign redundancy observed throughout Chapter 5). And fixed windows — as opposed to shot-aligned or content-adaptive ones — keep the ground-truth-to-segment mapping simple and the tIoU ceiling analysable (§6.2). Segment records are persisted as per-video JSONL and are the single interface between ingestion and everything downstream.

3.3 Embedding and dual-modality indexing

Each segment receives up to two vectors in a joint vision-language embedding space. The **visual vector** is the mean of its keyframes’ L2-normalised image embeddings, re-normalised — mean-pooling being the standard, hyperparameter-free aggregation that

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preserves the cosine geometry. The **text vector** embeds the segment’s transcript (plus OCR when enabled) with the *same* model’s text encoder. Using one joint space for both modalities is what allows a single encoded query to search either collection; the trade-off, accepted knowingly, is that a contrastive text encoder trained on alt-text is a weak reader of long or non-English transcripts (its 77-token context truncates long passages), which Chapter 5 quantifies.

The encoder itself sits behind a thin interface (`open_clip`) so that the backbone is a configuration value: ViT-B-32 for development, ViT-L-14 and SigLIP SO400M for the Chapter 5 ablations. Every backbone’s segment embeddings are cached as per-video arrays on disk, making a backbone swap a one-time re-embedding job and an index rebuild a matter of minutes — the property that made the W8 backbone comparison affordable.

Vectors live in ChromaDB — an embedded vector store chosen over server-class alternatives (Milvus, Qdrant) because at 10^3 – 10^4 segments HNSW [32] search is effectively exact and operational simplicity dominates. The two modalities are stored as **separate collections** sharing segment IDs and metadata (`video_id`, `start`, `end`, frame count) rather than as one concatenated vector. This is the single most consequential indexing decision: separate collections mean each modality can be searched, scored, and evaluated in isolation, which is precisely what RQ2’s ablations require; late fusion, early fusion, or either modality alone all become query-time choices over the same index. Segment text rides along as the collection’s document payload, so downstream components (re-rankers, agents) can access it without re-reading disk.

3.4 Retrieval

The retriever exposes one code path used identically by the evaluation harness, the agent, and the demo UI — deliberately, so measured numbers describe the system users interact with.

Single-query search encodes the query with the index backbone’s text encoder and performs cosine search in one modality, or in both with **late fusion** ($\text{score} = \alpha \cdot \text{visual} + (1-\alpha) \cdot \text{text}$, candidates over-fetched threefold before fusion). Late fusion was preferred over early (concatenation) fusion because it requires no joint training, is trivially ablatable per modality, and exposes α as an interpretable knob — the knob the α -sweep of §5.3.1

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then showed to have no useful setting on this corpus, itself a finding the design made cheap to obtain.

Query decomposition (W5) addresses the mismatch between causal/temporal question phrasing and CLIP’s training distribution of literal scene captions. An LLM rewrites the question into at most four short present-tense captions; each caption *plus the original question* is searched independently, and the rankings are combined by reciprocal rank fusion ($k = 60$). Two design details are load-bearing. Retaining the original question guarantees decomposition can only add candidates, never replace the baseline behaviour — a safety property that made it deployable by default. And rank-based fusion (rather than score averaging) sidesteps the incomparability of cosine scores across differently phrased sub-queries. Decompositions are cached on disk keyed by question text, so the marginal cost across the 3,358-question evaluation was a one-off ~US\$0.15.

Second-stage re-ranking (W6) re-scores a 30-candidate pool and re-orders it by a weighted RRF of the first-stage ranking and the re-ranker’s ranking. The design went through one documented failure: the planned text cross-encoder (question $\times$ transcript) *degraded* every metric, for reasons diagnosed in §5.3.3, and survives only as an ablation configuration. Its replacement re-scores candidates *visually*, with a stronger backbone (ViT-L-14) over pre-computed segment embeddings — turning the re-ranker into a second opinion from a better witness rather than a different modality. One methodological lesson from the failure is baked into both re-rankers: when a candidate lacks the re-ranker’s modality (no transcript; no cached vector), its re-ranker rank is *imputed from its retrieval rank* instead of being treated as absent, keeping all candidates on one score scale. Without this, every candidate the re-ranker could score would structurally outrank every candidate it could not — which in an early implementation silently drove R@1 to zero.

3.5 Agentic answering

The answering layer converts retrieved segments into a natural-language answer with per-claim citations of the form [video_id @ start-end s]. It is built as native LLM function-calling over two tools, in two harnesses of increasing sophistication.

Tools. `search_video_segments(query, modality)` wraps the retriever; its description instructs the model to phrase queries as scene descriptions rather than questions — pushing the decomposition insight into the agent’s tool-use behaviour. `get_segments_around(video_id,`

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`timestamp`, `direction`, `n`) is the temporal primitive: it returns the segments immediately before or after a timestamp in one video. Temporal questions (“what did X do *after* Y?”) are answered by composition — anchor Y with search, then walk **after** — a two-step plan the system prompt states explicitly, and which the agents were observed to follow (§5.4.1).

Simple agent (W4). A bounded tool-use loop: the model may call tools for up to three rounds, after which a final call with tool choice disabled forces an answer from the evidence gathered. The forced-answer step exists because both LLM providers, left to themselves, will occasionally keep requesting searches until the round budget expires and return an empty answer — an early bug whose fix (an explicit “budget exhausted, answer now” turn) applies to both harnesses.

Graph agent (W7). The LangGraph state machine of Figure 3.2 adds two things a linear loop cannot express cleanly. First, an explicit **reflect node**: when the agent produces a draft answer, a separate LLM pass audits it against the *actually gathered* evidence — do the cited segments exist in the tool results? are the claims supported rather than invented? is the temporal relation to the anchor correct? — and either accepts it (terminating the graph) or returns it once with a critique appended, resuming the tool loop. Reflection is bounded (one revision) and budget-aware: a revision is only requested if enough tool rounds remain to act on the critique, a guard added after observing budget-exhausted revision loops. Second, the graph makes the control flow inspectable: every run yields a typed trace of rounds, tool calls, and reflections, which Chapter 5 uses for its qualitative analysis.

Provider abstraction and the evidence channel. Both agents run against either an OpenAI-compatible endpoint (DeepSeek; any local server such as Ollama) or the Anthropic API. The abstraction is not merely operational: the two providers differ in the *evidence channel*. Under the multimodal provider, tool results embed each segment’s middle keyframe as an image block alongside its transcript, capped at six images per result; under text-only providers the same renderer emits transcript and timestamps only. Because the tools, prompts, and retrieval are identical across providers, switching the provider is a controlled manipulation of what evidence the answerer can perceive — the design property that enables the evidence-channel experiment of §5.4.2 to be run as a true single-variable comparison.

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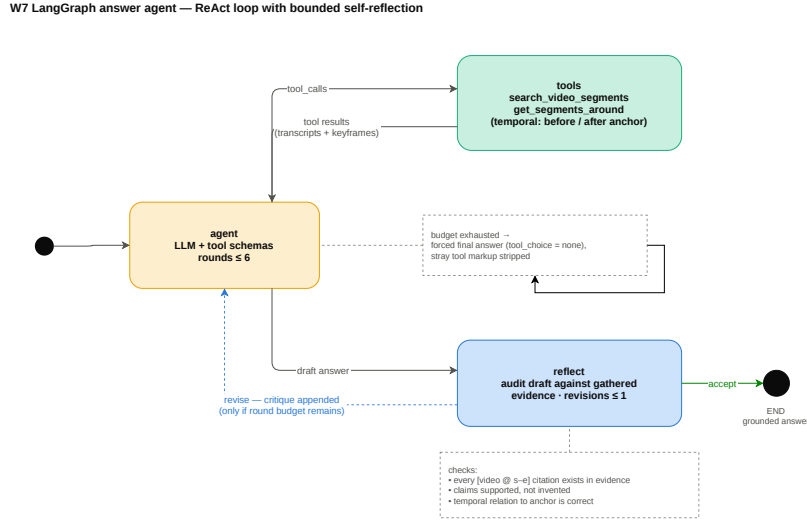


Figure 3.2: LangGraph answering-agent state machine with tools and bounded reflection.

3.6 Demonstration interface

A Streamlit application (Figure 3.3) exposes the full pipeline interactively in two tabs. The *search* tab runs the retrieval stack with per-component toggles — modality, query decomposition, visual re-ranking — and renders each hit with its keyframe, timestamps, transcript, and score; expanding a hit plays the source video seeked to the segment’s start time, closing the loop from natural-language query to the exact moment on screen. The *ask* tab runs either agent under either provider and displays the answer alongside its full evidence trail (every tool call and the segments it returned). The toggles map one-to-one onto the ablation dimensions of Chapter 5, which makes the interface an exploratory instrument as well as the project’s demonstrable artefact: behaviours noticed here motivated several of the diagnostics Chapter 5 then measured properly. The qualitative claims of Chapters 5 and 6 do not rest on it, however — those are drawn from the typed agent traces of §3.5, which record every tool call and reflection of a run.

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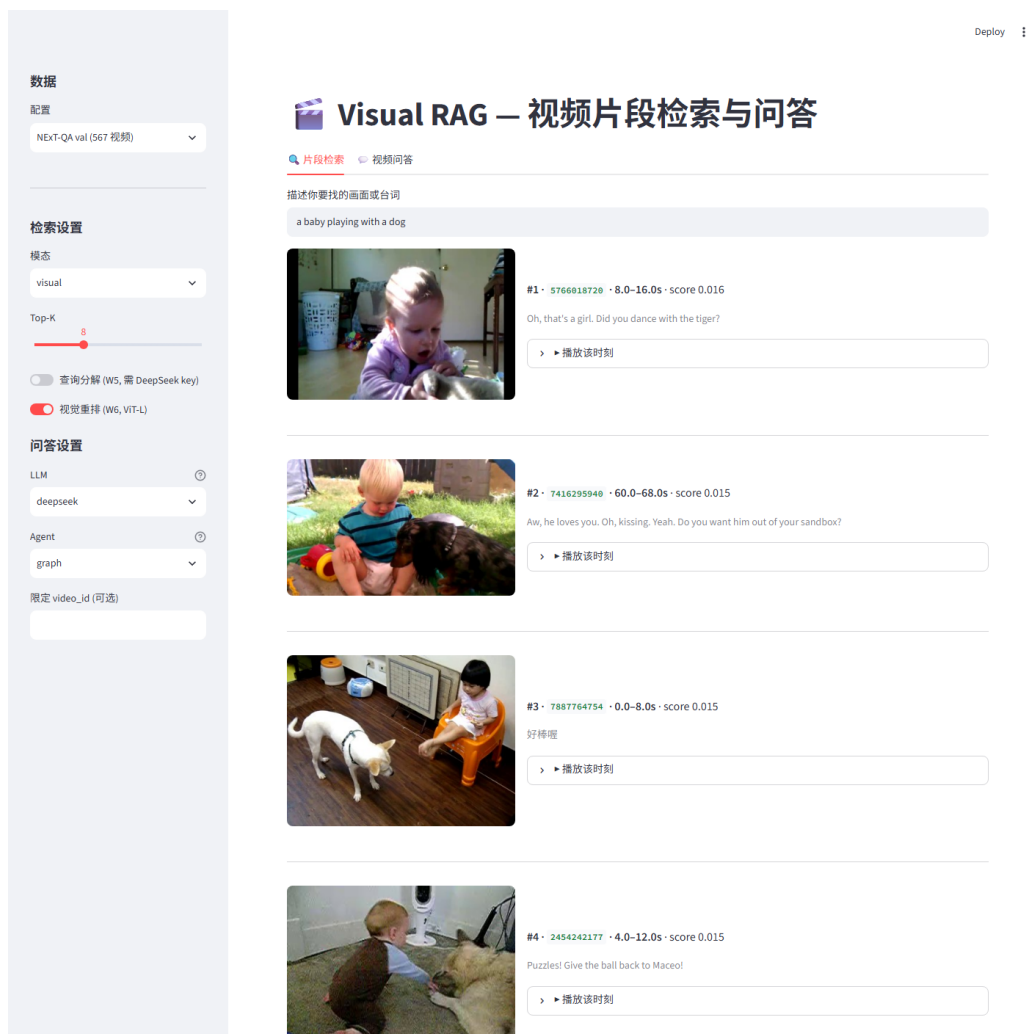


Figure 3.3: Visual RAG demonstration interface showing search results and timestamped evidence.

3.7 Summary

The design commits to cheap, measurable components over end-to-end training: fixed overlapping segments as the unit of temporal grounding; one joint embedding space with modalities indexed separately; retrieval improvements (decomposition, re-ranking) that

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compose as pure query-time re-rankings; an agent layer whose tools, control flow, and evidence channel are each independently switchable. Chapter 4 details the implementation of these components; Chapter 5 measures them.

4 Implementation

Chapter 3 described *what* was built and why; this chapter records *how*, with emphasis on the engineering that made a full benchmark evaluation feasible on a single consumer laptop (RTX 4050, 6 GB VRAM) and a few dollars of API usage. The codebase is a small Python package (`visualrag/`) with thin CLI entry points (`scripts/`), one YAML configuration per experimental condition, and a Streamlit interface (`ui/`).

4.1 Technology stack

Concern	Choice	Notes
Video decoding / duration	PyAV	single decode pass per video
Shot detection	PySceneDetect (adaptive)	+ 0.5 fps uniform fallback
Speech recognition	faster-whisper (large-v3, int8)	CTranslate2 runtime
Embeddings	open_clip: ViT-B-32, ViT-L-14, SigLIP SO400M	one interface, three backbones
Vector store	ChromaDB (embedded, HNSW, cosine)	two collections per backbone
Re-ranker (ablation)	sentence-transformers bge-reranker-v2-m3	text cross-encoder, §5.3.3
Agent framework	LangGraph	W7 state machine
LLM providers	DeepSeek, Anthropic Claude, any OpenAI-compatible endpoint	provider-agnostic answerer

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Concern	Choice	Notes
UI	Streamlit	search + timestamp-jump playback
Evaluation	custom harness (<code>visualrag/eval</code>)	all metrics of §5.1

Two stack decisions deserve one sentence each. ChromaDB was chosen over server-class vector databases because at 10^3 – 10^4 vectors, HNSW is effectively exact and zero operational overhead beats theoretical scalability. LangGraph was chosen over a heavier agent framework because the W7 design needed exactly one thing — an explicit, bounded, inspectable state machine — and nothing else.

4.2 Code organisation and the segment contract

The implementation is deliberately small: 2,162 lines across 19 modules in `visualrag/`, 1,019 lines of command-line entry points in `scripts/` (eleven of them), and a 234-line Streamlit application. The package’s seven sub-packages mirror the pipeline stages one-to-one — `ingest`, `embed`, `index`, `retrieve`, `agent`, `eval`, `data` — so the chapter structure of this dissertation and the directory structure of the repository are the same structure. Entry points hold no logic: each parses arguments, loads a configuration, and calls one library function, which is what allows the harness, the agent, and the UI to exercise identical code paths (§3.4).

Where the line counts concentrate is informative. `agent/answerer.py` (398 lines) and `agent/graph_agent.py` (268) dominate, followed by `retrieve/rerank.py` (171) and `retrieve/decompose.py` (152); every remaining module is under 130 lines. The asymmetry is the design working as intended — provider abstraction and control flow are genuinely irreducible complexity, while the pipeline stages stay thin because a single data contract does the coordinating between them.

The segment contract. That contract is the `Segment` dataclass in `schema.py`: an identifier, a video identifier, [`start`, `end`] seconds, keyframe paths, transcript text, and optional OCR text. Three properties of it carry more weight than its size suggests. First, the identifier is derived rather than allocated — `f"{video_id}::{start:.2f}-{end:.2f}"`

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— so it is reproducible from the timestamps alone; this is what lets per-backbone embedding caches, two ChromaDB collections, and the result JSONs be joined after the fact without a database or a migration step. Second, a `text` property concatenates whichever textual modalities are present, so retrieval code never branches on whether OCR is enabled; toggling that modality is a configuration change, not a code change. Third, segments persist as JSONL, which keeps every intermediate stage readable with ordinary tools and lets any stage be re-run from disk alone.

Configuration. Six YAML files describe the experimental conditions (`default`, `demo`, `ablation_vitl`, `ablation_siglip`, `agent_best`, `youcook2`), with command-line flags overriding individual fields. The point of this is provenance rather than convenience: a Chapter 5 run is identified by a file that is committed alongside its results, not by a line of shell history.

4.3 Fitting the pipeline into 6 GB

The plan’s premise (§3.1) was that heavy computation happens offline; the implementation’s contribution was making “offline” affordable *locally* rather than requiring a cloud GPU. Three techniques mattered.

Decoupling ASR from the deep-learning runtime. Whisper large-v3 nominally exceeds a 6 GB budget under PyTorch, but faster-whisper executes it through CTranslate2 with int8 weights in approximately 3 GB. One integration subtlety cost a bug fix: the pipeline’s device auto-detection originally probed *PyTorch*’s CUDA availability, which is false under a CPU-only PyTorch build even when the GPU is usable — so the transcriber now probes CTranslate2’s own CUDA device count, and ships the pip-installed cuBLAS/cuDNN libraries to the DLL search path on Windows. With this in place the full 567-video corpus transcribed in approximately 2 h locally.

Embedding caches as the unit of reuse. Every backbone’s segment embeddings persist as per-video compressed arrays (`.npz`) keyed by segment ID. Indexing reads only the cache; evaluation reads only the index. Re-running any experiment therefore costs seconds; adding a backbone costs one embedding pass; and the visual re-ranker (§3.4) is *free at query time* beyond a matrix product, because its ViT-L vectors are the same cache the ViT-L index ablation used. A related robustness detail: SigLIP’s `open_clip` implementation exposes no `visual.output_dim`, so the encoder wrapper resolves the

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embedding width by a three-step probe (attribute $\rightarrow$ model field $\rightarrow$ dummy forward pass) — the kind of small fix on which “one interface, three backbones” actually depends.

Idempotence everywhere. Every batch stage — ingestion, embedding, indexing, decomposition — checks for its own output and skips work already done, at per-video or per-question granularity. Long jobs on a laptop *will* be interrupted (sleep, OOM, session resets); with idempotent stages, every restart resumes where it stopped. During the project this was exercised repeatedly: the full-corpus ingestion survived two process kills and completed under a self-healing watchdog script that relaunches the batch until its completion marker appears (`scripts/ingest_watchdog.ps1`).

4.4 Cost engineering for LLM-dependent components

Two pipeline components call LLM APIs; both were engineered so that *evaluation scale* never multiplies *API cost*.

Query decomposition results are cached on disk keyed by the question string. The full validation split (3,358 questions, 16 concurrent workers) cost approximately US\$0.15 *once*; every subsequent experiment — the W5/W6/W8 ablations each rerun decomposition-dependent retrieval several times — replayed the cache for free. The cache is committed to the repository, making the retrieval experiments deterministic and reproducible without any API key.

Agent evaluation batches questions across a thread pool (the retrieval stack is read-only and thread-safe), parses the model’s final option choice from a fixed sentinel line, and records per-question token usage. The 150-question agent comparison cost approximately US\$0.35 (DeepSeek); the 44-question multimodal run approximately US\$3 and the 150-question multimodal best-configuration run approximately US\$25 (Claude, where keyframe image tokens dominate). Rate limits are respected by capping concurrency rather than by retry storms.

4.5 Provider-agnostic answering and its pitfalls

The answerer runs unchanged against Anthropic’s API, DeepSeek, or any OpenAI-compatible endpoint (including a local Ollama server — the hook for the open-source-model comparison left as future work). The abstraction is a thin dispatch: one message-

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format adapter per provider family, one shared tool schema, one shared evidence renderer that adds keyframe image blocks when the provider supports vision. Field experience contributed three fixes worth recording, since each silently produces *empty or corrupt answers* rather than errors. (i) Models sometimes spend the entire round budget requesting more searches; the loop must end with a forced-answer turn with tool use disabled. (ii) Under a forced-answer turn, DeepSeek occasionally emits its intended tool calls as literal markup in the message text; the answer extractor strips these. (iii) When self-reflection can send the agent back for more evidence, the revision path must check the remaining round budget, or the reflect/act cycle consumes it and the graph terminates answerless.

4.6 Evaluation harness and provenance

The harness (`scripts/evaluate.py`, `scripts/eval_qa.py`) exposes every experimental axis of Chapter 5 as a flag (`--scope`, `--modalities`, `--tau`, `--decompose`, `--rerank`, `--agent`, `--provider`) over per-condition YAML configurations, and emits a self-describing JSON per run. All result files are committed under `results/`; `docs/EXPERIMENTS.md` maps every number reported in this dissertation to the exact command, configuration, and date that produced it; and the dissertation’s data figures are generated from those JSONs by a committed script (`docs/figures/make_figures.py`), so no reported value exists only in prose. Figures 3.1–3.2 (architecture) are maintained as editable draw.io sources alongside their exports.

4.7 Correctness practices, and the absence of a test suite

The project has **no automated test suite**. This is stated plainly because the alternative practices it relied on are load-bearing for the credibility of Chapter 5, and because their gaps are real.

Four properties substituted for tests. *Determinism*: given the committed decomposition cache, a retrieval run is bit-identical across invocations, so re-running an unchanged configuration is itself a regression check and any metric movement is attributable to the change made. *Self-describing outputs*: each run emits a JSON recording its own configuration alongside its metrics, so a result cannot be silently mismatched with the settings that produced it. *Generated figures*: every data figure is produced from those

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committed JSONs by a script (§4.6), which makes it impossible for a figure and its prose to drift apart. *Idempotence*: a stage that finds its own output and skips it (§4.3) is an implicit consistency check on that output’s existence and shape.

In practice these caught defects at the level of implausible *numbers* rather than failing assertions. The re-ranker’s missing-modality bug (§3.4, §5.3.3) announced itself as an R@1 of exactly zero — a value no correct implementation could produce — and the ASR device-detection bug appeared as a silent fallback to CPU-speed transcription. Both were found because a monitored quantity moved, not because a test failed.

The limitation is the converse: this regime cannot detect a logic error in any path that no reported number exercises. The OCR ingestion path, disabled throughout (§3.2), is the clearest example — it is implemented, never measured, and therefore unverified. Unit tests over the segmenter’s boundary arithmetic and over the rank-fusion and imputation logic would be the highest-value addition to the codebase, and are recommended in §7.2 accordingly.

4.8 Summary

The implementation’s through-line is *reuse*: one decode pass per video, one embedding pass per backbone, one LLM call per unique question, one code path shared by harness, agent, and UI. That discipline — more than any individual optimisation — is what let a dissertation-scale evaluation programme (twenty-odd full-split retrieval runs, three agent studies, three backbones) execute on one laptop within a single project week.

5 Evaluation

This chapter evaluates the system against the three research questions. Section 5.1 describes the experimental setup shared by all experiments. Section 5.2 addresses RQ1 by comparing multi-modal retrieval against a text-only baseline. Section 5.3 addresses RQ2 through a series of ablations over the embedding and retrieval strategy — modality fusion, query decomposition, second-stage re-ranking, and the embedding backbone — including two negative results that shaped the final design. Section 5.4 addresses RQ3 by measuring end-to-end question-answering accuracy of the agentic pipeline, culminating in a controlled experiment that isolates the contribution of visual evidence at the answering stage. Section 5.5 replicates the modality comparison on a speech-dense second domain (YouCook2), testing the domain-conditionality prediction that Chapter 6 attaches to the text-channel negative results. Section 5.6 reports system-level costs. Every number in this chapter is reproducible from the committed result files (`results/*.json`); the exact command that produced each result is recorded in `docs/EXPERIMENTS.md`.

5.1 Experimental setup

Dataset. All experiments use the validation split of NExT-QA [11], a benchmark of causal and temporal multiple-choice questions over short (~44 s) everyday videos, together with the temporal grounding labels contributed by NExT-GQA [12]. The split contains 3,358 questions over 567 videos, and — after intersecting with the grounding annotations — every question carries at least one ground-truth temporal interval locating the evidence for its answer. NExT-GQA was chosen over larger retrieval benchmarks (e.g. MSRVT, ActivityNet Captions) precisely because these interval labels allow retrieval to be scored with *temporal* precision rather than at whole-video granularity.

Indexing. The ingestion pipeline of Chapter 3 produced 5,725 overlapping segments (8 s window, 4 s stride) from the 567 videos. Each segment is indexed in two ChromaDB

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collections (HNSW, cosine): a visual collection holding the pooled CLIP embedding of the segment’s keyframes (5,725 vectors), and a text collection holding the CLIP text embedding of its speech transcript (3,433 vectors — 60% of segments contain speech). Unless stated otherwise the index backbone is CLIP ViT-B-32 (laion2b); Section 5.3.4 varies the backbone.

Retrieval metrics. For each question, the question text is used as the query and the top-K segments are retrieved. A retrieved segment counts as a *hit* if it comes from the ground-truth video **and** its temporal interval overlaps a ground-truth interval with $\text{tIoU} \geq \tau$; requiring both conditions ties the metric to moment-level retrieval rather than video identification. Reported metrics are $\text{Recall@}\{1,5,10\}$ and MRR at the primary threshold $\tau = 0.5$, nDCG@10 with graded tIoU gains (ideal-normalised against the ground-truth video’s own segments), and tIoU@1 — the mean temporal overlap of the top-ranked segment, the dissertation’s “temporal accuracy” measure. Section 5.3.5 repeats key configurations at the looser $\tau = 0.3$.

Scopes. Two retrieval scopes are evaluated. In **corpus** scope the query searches all 5,725 segments across 567 videos — the full RQ1 setting, where the system must find both the right video and the right moment. In **video** scope the search is restricted to the ground-truth video’s own segments, isolating pure temporal localisation. The gap between the two scopes attributes error to cross-video confusion versus within-video imprecision.

QA metric. Section 5.4 evaluates answer generation directly on NExT-QA’s five-way multiple-choice format: the agent receives the question and its five options, gathers evidence with its tools, and must emit a final option index. Accuracy is reported overall and per question type (CW/CH: causal *why/how*; TN/TC/TP: temporal *next/current/previous*). The random-guess floor is 0.20.

Implementation constants. Unless varied by the ablation, fusion weight $\alpha = 0.5$, retrieval depth $K = 10$, re-ranker candidate pool 30, and the question encoder matches the index backbone. All retrieval experiments are deterministic given the committed decomposition cache; agent experiments use temperature-default API calls and are therefore subject to normal LLM variance.

5.2 RQ1 — Multi-modal retrieval against a text-only baseline

Table 5.1 reports the three single-strategy configurations on both scopes at $\tau = 0.5$. The text-only baseline embeds each segment’s Whisper transcript and searches it with the question text — the standard “search what was said” approach that RQ1 asks the system to beat. A reminder of scale before reading the table (§1.2): a hit requires the correct video *and* $\text{tIoU} \geq 0.5$ between an 8-second window and the annotated moment, searched over all 5,725 segments — a deliberately strict criterion under which absolute scores are small by construction, so the informative content of the table is in the *relative* comparisons between rows, not in the magnitudes.

Table 5.1 — Retrieval by modality (ViT-B-32 index, $\tau = 0.5$).

Scope	Modality	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
corpus	visual	.026	.075	.111	.047	.117	.035
corpus	text	.002	.005	.011	.004	.011	.003
corpus	fused ($\alpha=.5$)	.002	.010	.013	.005	.018	.007
video	visual	.148	.394	.491	.250	.659	.203
video	text	.080	.232	.305	.143	.390	.117
video	fused ($\alpha=.5$)	.129	.384	.491	.235	.639	.186

Three observations answer RQ1. First, *visual retrieval dominates text retrieval by an order of magnitude in corpus scope* (R@10 .111 vs .011): to find a moment among 567 videos of everyday footage, what is *on screen* is far more discriminative than what is *said*. Second, two causes plausibly account for the failure of the text channel, and neither is poor transcription as such. NExT-QA’s home videos contain sparse, conversational, and frequently non-English speech (German, Mandarin and other languages appear in the transcripts), so the information a question asks about is often simply *absent* from the speech — a question about what a child *does* is rarely answered by what anyone *says*. And the CLIP text encoder that embeds both queries and transcripts is trained predominantly on English alt-text, so even present information may be *unreadable* to it. These two causes are entangled in this experiment; §6.2 explains why the present design cannot apportion blame between them, and names the translate-then-embed variant that

5 Evaluation

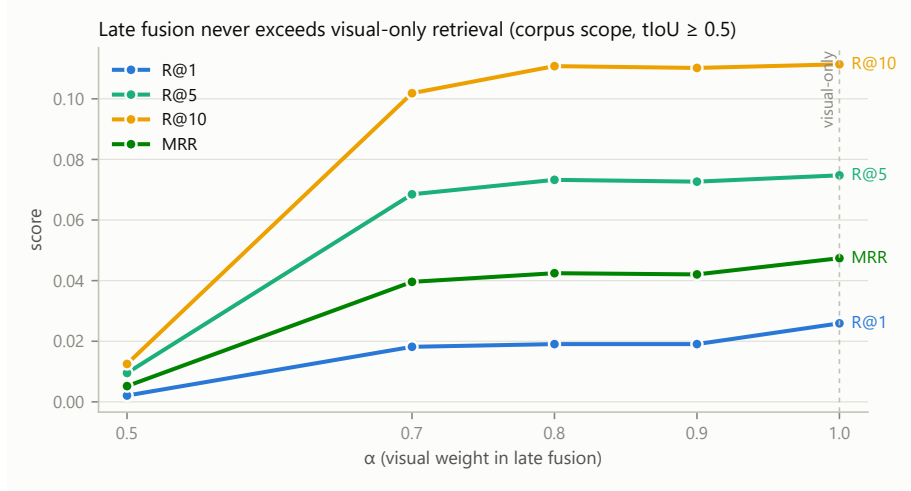


Figure 5.1: Late-fusion weight sweep in corpus scope.

would. Third, the corpus/video gap localises the difficulty: within the correct video the system already ranks a correct moment in its top five for 39% of questions, but across the corpus R@10 falls to 11% — *cross-video confusion, not within-video imprecision, is the dominant error mode*, which motivates the precision-oriented interventions of Section 5.3.

The by-type breakdown (Appendix B) shows temporal-precise question types (TP, $n = 52$) retrieving worst in corpus scope, previewing the temporal-reasoning theme of Section 5.4.

5.3 RQ2 — Embedding and retrieval strategy ablations

5.3.1 Negative result: naive late fusion never beats visual-only

The natural response to a weak-but-nonzero text channel is weighted late fusion: $\text{score} = \alpha \cdot \text{visual} + (1-\alpha) \cdot \text{text}$. Figure 5.1 sweeps α over $\{0.5, 0.7, 0.8, 0.9\}$ in corpus scope. Every metric rises monotonically toward the $\alpha = 1.0$ endpoint — visual-only — and none crosses it; the same holds in video scope, where the best fused configuration ($\alpha = 0.8$) ties visual-only on R@10 (.494 vs .491) but loses on R@1 (.139 vs .148) and MRR (.244 vs .250). At $\alpha = 0.5$ fusion is catastrophic in corpus scope (R@10 .013) because the text scores act as high-variance noise added to a weak-but-real visual signal.

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The conclusion is sharper than “tune α ”: *with this text channel, there is no fusion weight at which the transcript embeddings contribute positive evidence*. Fusion is not a weighting problem but a channel-quality problem — the actionable implication being to repair the text channel (e.g. translate transcripts to English before embedding, or use a dedicated multilingual text-retrieval encoder) before re-introducing it, rather than to search the weight space further. All subsequent experiments therefore use the visual modality.

5.3.2 Query decomposition recalls more, but no more precisely

NExT-QA questions are causal and temporal (“why did the boy pick up one present from the group of them and move to the sofa”), whereas CLIP matches literal scene descriptions. The W5 decomposer prompts an LLM to rewrite each question into at most four short scene captions (“a boy picks up a present from a group of presents”, “a boy carries a present and walks to a sofa”, ...), retrieves each caption plus the original question independently, and fuses the rankings with reciprocal rank fusion. The rewrite is computed once per question and cached (3,358 questions approximately US\$0.15 of LLM usage).

Decomposition lifts the deep-rank metrics in corpus scope — $R@10 .111 \rightarrow .122$ (+9.9% relative), $R@5 +6.7\%$, $MRR +6.4\%$ — while leaving the top of the ranking essentially untouched ($R@1 .026 \rightarrow .025$, $tIoU@1$ unchanged at .035; Table 5.2, rows 1–2; Figure 5.2). The pattern is informative: translating abstract questions into CLIP-native captions surfaces *additional correct candidates* into the top ten, but rank fusion of near-tied cosine scores cannot push them to position one. Recall problems and precision problems are separable; decomposition solves a recall problem, and the next section supplies the missing precision.

5.3.3 Second-stage re-ranking: a negative and a positive result

Text cross-encoder (negative). The project plan anticipated a standard text cross-encoder re-ranker (bge-reranker) scoring (question, transcript) pairs over the candidate pool. On a 100-question development subset this *reduced* quality at every fusion weight tested — at weight 0.5 it halved $R@5$ (.070 $\rightarrow$.030), and even at 0.8 it remained below the no-reranking baseline. One asymmetry should be stated plainly: this negative

5 Evaluation

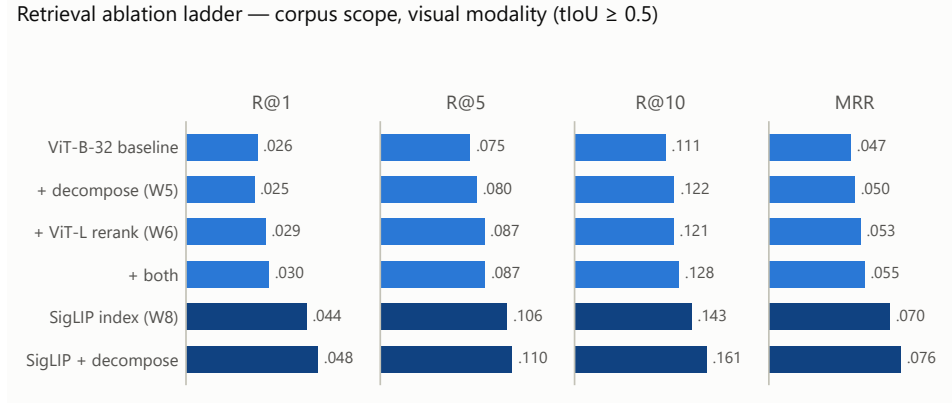


Figure 5.2: Retrieval ablation ladder from the baseline to the final configuration.

result rests on 100 questions, whereas every positive retrieval result in this chapter is measured on all 3,358. The component was rejected on the development subset and never promoted to a full-split run, so the evidence against it is weaker than the evidence for the components that replaced it — a decision revisited in §6.3. The mechanism mirrors §5.3.1: the transcripts are conversational chatter, largely uncorrelated with the visual events the questions ask about, so every reordering the cross-encoder makes injects noise. An implementation subtlety compounds the problem and is worth recording: 40% of segments have no transcript at all, and if such candidates simply receive no re-ranker score, any scored candidate the cross-encoder likes outranks *every* unscored one — structurally demoting exactly the visual-only segments most likely to be correct. (The first implementation here had this flaw, and drove R@1 to zero.) The repaired design imputes the re-ranker term of a modality-missing candidate from its retrieval rank, keeping both candidate classes on one score scale; the negative result above is reported *with* this repair, so it reflects the channel, not the bug.

Visual re-ranker (positive). The replacement re-ranker re-scores the top-30 candidates with a stronger vision model (CLIP ViT-L-14) over pre-computed segment embeddings, fusing the ViT-L ranking with the first-stage ranking by weighted RRF. Every segment has keyframes, so the missing-modality problem vanishes. Table 5.2 and Figure 5.2 show the ladder at $\tau = 0.5$, corpus scope:

Table 5.2 — Corpus-scope ablation ladder (visual modality, $\tau = 0.5$).

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Configuration	R@1	R@5	R@10	MRR	tIoU@1
ViT-B-32 baseline	.026	.075	.111	.047	.035
+ decompo- sition (W5)	.025	.080	.122	.050	.035
+ ViT-L re-ranking (W6)	.029	.087	.121	.053	.041
+ both	.030	.087	.128	.055	.040
SigLIP index (W8)	.044	.106	.143	.070	.054
SigLIP + de- composition	.048	.110	.161	.076	.055

The two mid-pipeline interventions compose additively: decomposition contributes at depth, re-ranking at the top of the list, and their combination improves every metric over the baseline (+15–17% relative on MRR and R@10). In video scope the combined configuration reaches R@1 .154, R@10 .501, tIoU@1 .209 — each the best of any ViT-B configuration.

5.3.4 Backbone: SigLIP dominates, and two-stage retrieval matches one-stage

The final RQ2 axis exchanges the embedding backbone itself, holding segmentation and evaluation fixed. Three backbones were compared as *the index*: CLIP ViT-B-32 (baseline), CLIP ViT-L-14, and SigLIP SO400M (webli), each with its own question encoder.

Two findings emerge from Table 5.2 (bottom rows) and Table 5.3. First, *SigLIP is categorically stronger for this task*: as the index it lifts corpus R@1 by 69% relative over ViT-B (.026 $\rightarrow$.044) and tIoU@1 by 54%, and with decomposition on top reaches R@1 .048 — *nearly double the original baseline* — and video-scope tIoU@1 .230. This is consistent with SigLIP’s sigmoid-loss pretraining, which has been reported to favour retrieval tasks [4].

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Second, ViT-L used as the full index (corpus R@1 .029, R@5 .090, R@10 .122, MRR .054) is indistinguishable at the resolution of this evaluation from ViT-B *plus* ViT-L re-ranking (.029/.087/.121/.053) — every metric agrees to within ± 0.003 , though the pair was compared numerically and not by a significance test. *The two-stage cheap-index/expensive-re-ranker design recovers essentially all of the larger model’s quality* while embedding the corpus only with the small model; at larger corpus scales, where re-embedding everything with the expensive model is prohibitive, the two-stage design is the scalable route to the same accuracy. A corollary worth stating for practitioners: a re-ranker must *differ* from the index model — re-scoring with the index’s own embeddings is an identity operation.

Table 5.3 — Video-scope backbone comparison (visual modality, $\tau = 0.5$).

Index					
backbone	R@1	R@5	R@10	MRR	tIoU@1
ViT-B-32	.148	.394	.491	.250	.203
ViT-L-14	.151	.407	.496	.257	.210
SigLIP	.170	.406	.494	.266	.221
SO400M					
SigLIP + de-composition	.176	.419	.496	.275	.230

5.3.5 Threshold sensitivity

$\tau = 0.5$ is a strict criterion for 8-second windows scored against ground-truth intervals of arbitrary length. Table 5.4 repeats the ViT-B baseline and the best ViT-B configuration at $\tau = 0.3$; the binary metrics roughly double, while tIoU@1, which is threshold-free, is unchanged by construction.

Table 5.4 — Threshold sensitivity (visual modality, ViT-B index).

Scope	Configuration	τ	R@1	R@5	R@10	MRR
corpus	baseline	0.5	.026	.075	.111	.047
corpus	baseline	0.3	.054	.130	.186	.089
corpus	+ decompose + rerank	0.5	.030	.087	.128	.055

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Scope	Configuration	τ	R@1	R@5	R@10	MRR
corpus	+ decompose + rerank	0.3	.062	.151	.208	.101
video	baseline	0.5	.148	.394	.491	.250
video	baseline	0.3	.311	.660	.774	.456
video	+ decompose + rerank	0.5	.154	.410	.501	.258
video	+ decompose + rerank	0.3	.323	.677	.780	.469

Two conclusions follow: the system’s practical localisation ability (“lands within a loose overlap of the right moment”) is considerably better than the strict headline numbers suggest — within the correct video, a correct moment is in the top five for two-thirds of questions — and the relative ordering of configurations is stable across thresholds, so the ablation conclusions above do not depend on the choice of τ .

5.4 RQ3 — Agentic question answering

5.4.1 Agent comparison

Table 5.5 and Figure 5.3 compare the two agent designs of Chapter 3 on the first 150 grounded val questions under a text-only LLM (deepseek-chat [33]): the single-loop function-calling agent (W4) and the LangGraph state machine (W7), which adds the temporal tool (`get_segments_around`) and a bounded self-reflection pass that audits the draft answer’s citations against the gathered evidence before accepting it.

Table 5.5 — Five-choice QA accuracy, 150 val questions. Three variables move across these rows — agent, base model, and evidence channel — so only rows sharing two of the three are directly comparable; §5.4.3 draws the valid pairings. TP is omitted from the by-type columns ($n = 2$) and reported in Appendix B.

	Base			CW	TN	TC	CH
Agent	model	Evidence	Overall	(n=61)	(n=44)	(n=22)	(n=21)
Simple loop (W4)	deepseek- chat	text- only	.447	.459	.341	.591	.429

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Agent	Base model	Evidence	Overall	CW (n=61)	TN (n=44)	TC (n=22)	CH (n=21)
LangGraphdeepseek- (W7)	chat	text- only	.547	.590	.341	.773	.619
none — prior only (§5.4.3)	deepseek- chat	none	.560	.574	.568	.545	.524
none — prior only (§5.4.3)	claude- opus-4-8	none	.560	.639	.455	.591	.476
LangGraphclaude- (W7), best config (§5.4.4)	opus-4-8	multimodal	.727	.852	.568	.682	.762

Both agents clear the .20 random floor by a wide margin — although §5.4.3 will show that the random floor is the wrong bar, since the bare model answers at .560 with no evidence at all; the informative comparison in this section is between the two agents, which saw identical evidence. The graph agent improves overall accuracy by 10 points (+22% relative), with the gains concentrated in causal questions (CW +13.1, CH +19.0 points) and temporal-current questions (TC +18.2). Because both agents answered the identical 150 questions, the comparison is paired: 29 questions flipped to correct under the graph agent against 14 that flipped to incorrect, giving an exact McNemar $p = .031$, with a paired-bootstrap 95% CI on the difference of $[+.013, +.187]$ (10,000 resamples) — the improvement is unlikely to be run-to-run LLM variance. Inspection of transcripts attributes the gains to three behaviours: systematic anchor-then-walk retrieval on temporal questions, reformulated follow-up searches when initial evidence is thin, and the reflection pass — which in observed runs rejected drafts whose citations did not appear in the gathered evidence, forcing a grounded rewrite.

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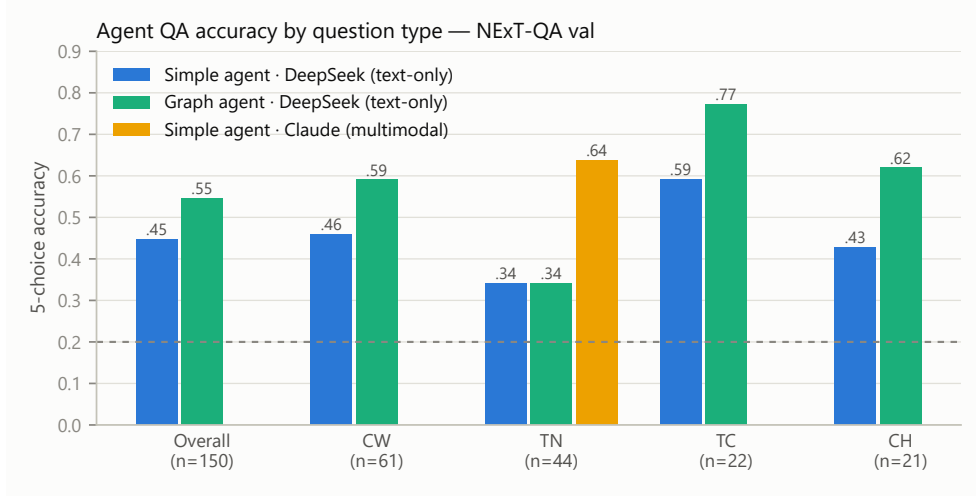


Figure 5.3: Question-answering accuracy by question type and agent configuration.

The striking exception is TN (*what happened next*), frozen at .341 for **both** agents. Qualitative inspection shows the temporal machinery working exactly as designed — the agent anchors the cue event and retrieves the segments that follow it — and then failing at the last step: the answer to a TN question is almost always a *visual action* (what a person or animal does next), the following segments are typically silent, and a text-only LLM receives no usable evidence. In one representative case (“what does the white dog do after going to the cushion”, ground truth *smells the black dog*), the text-only agent responded that the post-anchor segments contain no speech and the action “cannot be determined from the available evidence” — a correct description of its own evidence starvation.

5.4.2 Controlled experiment: visual evidence at the answering stage

The TN result yields a precise hypothesis: the bottleneck is the *evidence channel*, not the temporal mechanism. To test it, the same 44 TN questions were re-run under the *simple* agent with a single change from its own text-only row: the provider was switched to a multimodal LLM (claude-opus-4-8), whose tool results embed each segment’s keyframe image alongside the transcript. Retrieval index, tools, prompts, and questions were held fixed. The agent is named because it matters for the comparison: the LangGraph harness

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targeted OpenAI-compatible providers only at the time of this experiment, so the strict single-variable contrast is row 1 against row 3 of Table 5.6. The graph-agent row is listed because it lands on the identical .341, which makes the same conclusion hold against it; the graph-plus-multimodal combination was implemented later and is evaluated in §5.4.4.

Table 5.6 — Same 44 TN questions, evidence channel varied.

Configuration	TN accuracy
Text-only LLM, simple agent	.341
Text-only LLM, graph agent (temporal tool)	.341
Multimodal LLM, simple agent, keyframes in evidence	.636

Accuracy rises by 29.5 points (+87% relative, 44/44 questions completed). This comparison is likewise paired on identical questions: against the graph agent, 16 questions flipped to correct under visual evidence and 3 flipped away (exact McNemar $p = .004$; paired-bootstrap 95% CI $[+.114, +.455]$; against the simple agent, $p = .007$). On the representative case above, the multimodal agent answered “leans down and sniffs/nuzzles the small black puppy [28–36 s]” — matching the ground truth almost verbatim from the keyframes alone; Figure 5.4 shows the keyframes in question beside the two models’ verbatim answers. Because everything except the evidence modality was held constant, the improvement is attributable to visual evidence at the answering stage; §5.4.3 adds the absolute calibration — .636 also clears the multimodal model’s own no-evidence prior on TN (.455) by +.18. Together with Section 5.2, this closes the loop on the modality question at *both* ends of the pipeline: retrieval needs visual embeddings to find the right moment, and generation needs visual evidence to describe it.

5.4.3 Prior-only baselines: what the .341 floor actually was

Two readings of the text-only TN floor deserve separation. Part of the margin above random could reflect genuine transcript evidence; part could reflect NExT-QA answer priors — a language model can sometimes reject implausible distractors without any evidence. To separate them, both models answered the same 150 questions with *no tool access at all* — the question and its five options, nothing else (Table 5.5, prior-only rows; 150/150 completed each).

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"What does the white dog do after going to the cushion?"
Same retrieval, same tools — only the evidence channel differs



Text-only LLM — evidence starved

"...after going to the cushion, the video simply continues in silence — no further spoken action is described. To know exactly what the dog physically does (e.g., lies down, sniffs, sits), the keyframe images would need to be inspected."

Multimodal LLM — matches ground truth ("smells the black dog")

"...the white dog leans down and sniffs/nuzzles the small black puppy that is lying in the cushion [@ 28–36 s], then pushes its nose into the bedding toward the puppy [@ 36–44 s]."

Figure 5.4: White-dog qualitative case: retrieved keyframes and answers under text-only and visual evidence.

The result overturns the natural reading of the floor. Both models score .560 overall without evidence — coincidentally identical headlines over different by-type profiles — and the text-only model's own TN prior is **.568**. The .341 text-only TN figure was therefore never a floor: it is a *suppression*. Given transcript-only evidence on questions whose answers are visual, the same model performs .227 **below** its own prior (paired on the 44 TN questions: 14 flips away from correct against 4 toward it; exact McNemar $p = .031$, bootstrap 95% CI $[-.409, -.045]$). The mechanism is visible in the §5.4.1 transcripts: instructed to ground its answer in evidence, the model dutifully reasons from segments that contain nothing relevant — and abandons the prior that would have served it better. When the channel carries no signal for the question asked, delivered evidence is not merely unhelpful; it is anti-informative.

The same accounting reframes the agent comparison. The simple agent's .447 sits significantly below the bare prior ($p = .024$, CI $[-.207, -.020]$); the graph agent's .547

recovers only to parity with it ($p = .88$). And the recovery is not uniform but a sum of opposing type-level effects: transcript evidence genuinely beats the prior where speech describes the concurrent scene (TC .773 vs a .545 prior), is roughly neutral on the causal types (CW .590 vs .574; CH .619 vs .524), and is destructive on TN as above. Averaged over the type mix, the best text-only stack lands where the bare model started.

Visual evidence is the only channel measured that beats the prior. Within the multimodal model, the full stack adds +.167 over its own prior (.560 $\rightarrow$.727; 37 flips up against 12 down; exact McNemar $p = .0005$, CI [+0.080, +.253]), and on TN the keyframe evidence of §5.4.2 lifts its .455 prior to .636. Three consequences follow. First, the .200 random floor is the wrong bar for multiple-choice video QA — the right bar is the bare model’s prior, and every absolute accuracy in this chapter should be calibrated against .560. Second, the clean attributions are within-model, within-channel pairs; comparisons across Table 5.5’s rows conflate base-model strength, evidence channel, and agent machinery. Third, grounding discipline has a measurable cost when the channel is empty — a design implication taken up in §6.1.

5.4.4 Best-configuration run and contextualisation with published systems

The strongest components identified across this chapter were finally combined into one configuration: the SigLIP index (§5.3.4) with ViT-L visual re-ranking inside the agent’s search tool, the LangGraph agent (§5.4.1), and the multimodal evidence channel (§5.4.2). On the same 150-question sample this configuration reaches **.727** overall (Table 5.5, bottom row; 95% CI [.655, .798]; zero failed runs; approximately US\$25 of API usage). The gains concentrate where each component predicts: causal questions reach .852 (CW) and .762 (CH) — the multimodal evidence lets the model *see* the causes it previously had to guess — and TN rises from .341 to .568, confirming §5.4.2’s diagnosis while remaining the hardest type.

One result inside that figure runs against the trend and should be stated rather than averaged away. On the same 44 TN items, under the same multimodal evidence channel, the graph agent’s .568 is *three questions below* the simple agent’s .636 (§5.4.2): adding the temporal tool and the reflection pass did not help here, and marginally hurt. At $n = 44$ a three-question gap is well inside sampling noise and no claim of a real regression

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is made. The direction is nonetheless consistent with a structural gap in the current design — the reflection node audits a *text* rendering of the gathered evidence, so on questions whose answer is a silent visual action it interposes a text-mediated check on a judgement the model made from keyframes, and has nothing to contribute. This makes the keyframe-aware reflection pass the one component interaction the dissertation leaves untested (§6.2), and the headline TN figure for the evidence-channel effect remains §5.4.2’s .636 under the simple agent.

To situate these numbers, Table 5.7 lists published zero-shot results on the NExT-QA validation set alongside this system’s, and the closest published temporal grounding figures on NExT-GQA.

Table 5.7 — Contextualisation with published systems (protocols differ; see caveats below).

System	Protocol	NExT-QA val acc.
Random	—	.200
SeViLA zero-shot [13]	full val, fine-tuned localiser	approximately .636
LLoVi [27]	full val, GPT-based	approximately .677
VideoAgent [28]	full val, GPT-4 + iterative frame selection	.713
This work, best config	grounded 150-q sample, ≤ 6 segments/search	.727 [.655, .798]

Grounding	Protocol	metric
FrozenBiLM (NG+) [12]	NExT-GQA test, answer grounding	mIoU .096
Temp[CLIP] (NG+) [12]	NExT-GQA test, answer grounding	mIoU .121
SeViLA [12]	NExT-GQA test, answer grounding	mIoU .217
This work, SigLIP + decomposition	NExT-GQA val, top-1 retrieved segment	tIoU@1 .230

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Three caveats govern the reading of Table 5.7, and are repeated wherever these numbers are cited. First, the QA comparison is *indicative, not a leaderboard claim*: published systems evaluate the full approximately 5,000-question validation set while the figure here comes from a 150-question sample of the grounded subset, so the honest statement is that the best configuration performs *at, and in point estimate above, the published zero-shot state of the art on this sample*, with a confidence interval that overlaps the strongest published figure. Second, the comparison is not equal-resource: this system answers from at most six retrieved segments per search under approximately US\$0.17 per question, but benefits from a stronger underlying LLM (claude-opus-4-8) than the GPT-4-era published systems — LLM generation and method both differ, and this chapter’s internal ablations (not Table 5.7) are what isolate the method’s contribution. The prior-only baseline (§5.4.3) quantifies the model-strength component directly: claude-opus-4-8 reaches .560 on these questions with no tools at all, so the honest within-model attribution for the method is $.560 \rightarrow .727 (+.167, p = .0005)$. Third, the grounding rows compare related but non-identical protocols (their mIoU grounds a QA model’s answer on the test split; tIoU@1 here scores the top retrieved segment on val); they are included because both measure “where the system points for its evidence”, where this retrieval-first design proves competitive with — and in point estimate above — published weakly-supervised grounding, while additionally emitting the citation as a first-class output.

5.5 Cross-domain replication: speech-dense video

The causal analyses of §5.2 and §5.3.1 blame the text channel’s failure on a property of the *domain* rather than of the components: NExT-QA’s home videos are speech-sparse, 60% of segments carrying any speech and much of that multilingual chatter. That reading makes a falsifiable prediction — in a speech-dense, single-language domain the same components should reverse sign — and this section tests it directly.

Setup. Forty validation videos of YouCook2 [34], which is English-narrated cooking instruction of mean duration approximately 5 minutes, were ingested by the *unchanged* pipeline: the same 8 s / 4 s segmentation, the same Whisper transcription, and the same ViT-B-32 index as the §5.2 baseline, producing 3,110 visual and 3,100 transcript vectors over 3,173 segments. The benchmark’s own step annotations serve as queries. Each annotated step (a sentence plus a ground-truth [start, end] interval) becomes one moment-

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retrieval query, 314 in total, scored under the §5.1 protocol. The speech-density contrast is stark: 97.7% of segments carry speech (3,100 of 3,173, against 60% on NExT-QA), and the speech is task-aligned narration rather than incidental conversation.

Table 5.8 — Modality comparison on YouCook2 (ViT-B-32 index, $\tau = 0.5$, 314 step queries).

Scope	Modality	R@1	R@5	R@10	MRR	tIoU@1
corpus	visual	.048	.099	.143	.076	.058
corpus	text	.096	.213	.261	.145	.128
corpus	fused ($\alpha=.5$)	.099	.197	.287	.145	.108
video	visual	.115	.245	.318	.173	.152
video	text	.146	.331	.392	.221	.199
video	fused ($\alpha=.5$)	.169	.350	.420	.247	.215

Every sign flips as predicted. Transcript retrieval **doubles** visual retrieval in corpus scope (R@1 .096 vs .048; MRR .145 vs .076) — the mirror image of §5.2, where visual led text by an order of magnitude. Late fusion, which on NExT-QA never beat visual-only at any weight (§5.3.1), is here the *best configuration in video scope on every metric* (MRR .247 against .221 text-only and .173 visual-only) and the best deep-rank configuration in corpus scope (R@10 .287). The α sweep confirms the direction of the regime: quality now *falls* as the visual weight rises (corpus R@10 .287 at $\alpha = .5 \rightarrow .194$ at $\alpha = .8$), where on NExT-QA the same sweep improved monotonically toward visual. The ordering is unchanged at the looser $\tau = 0.3$ (corpus text R@1 .191 vs visual .099).

The replication turns the conditionality *claim* of §5.3.1 into a measured result: those negative results are properties of the domain, not of the components.

Part of the reversal, however, is *register* rather than speech density. YouCook2 queries are step descriptions phrased in the narrator’s own vocabulary, where NExT-QA questions are causal reformulations of silent visual events. The two factors move together in this design and it cannot separate them; doing so would need a third condition, speech-dense video queried out of register, which was not run. What the replication does establish is the weaker but sufficient claim: with every component held fixed, changing the domain reverses the ordering, so that ordering is not a property of the components. The practical corollary is a *domain-adaptive* modality policy, weighting the transcript channel by

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measured speech density and register match with exactly this harness, rather than a fixed design choice.

Two qualifications temper the result. The second corpus is small (40 videos, sized to the remaining project budget), and the replication covers the retrieval layer only, since the QA-stage experiments of §5.4 were not repeated. The direction and magnitude of the reversals ($2\times$ on corpus R@1, fusion best across the board in video scope) are nonetheless far outside plausible noise for a deterministic 314-query evaluation.

5.6 System-level measurements

The offline stage processed all 567 videos in approximately 2 h on a single RTX 4050 laptop GPU (6 GB): keyframe extraction at <1 s per video and Whisper large-v3 (int8, CTranslate2) transcription at 2–20 s per video; CLIP ViT-B-32 embedding and indexing of 5,725 segments adds approximately 15 min, and the SigLIP index approximately 35 min. The resulting index is portable (17 MB of vectors per backbone in npz form plus the ChromaDB store) and queries run interactively: question encoding is sub-second on GPU, HNSW search over 5,725 vectors is milliseconds, and the visual re-ranker adds one batched matrix product over 30 candidates. End-to-end agent answers complete in 15–60 s depending on tool rounds, dominated by LLM latency. Marginal API costs are small: query decomposition for the entire split cost approximately US\$0.15 (cached thereafter), a 150-question text-only QA run approximately US\$0.35, and the 44-question multimodal run approximately US\$3. The retrieval programme and the text-only QA studies — every experiment of §5.2, §5.3 and §5.5, plus the agent comparison of §5.4.1 — are therefore reproducible on one consumer laptop plus approximately US\$5 of API usage. The one expensive run is the 150-question multimodal best configuration of §5.4.4, at approximately US\$25 (approximately US\$0.17 per question, dominated by keyframe image tokens), bringing the full chapter to approximately US\$30 end to end.

5.7 Summary

RQ1. Visual retrieval outperforms the text-only baseline by an order of magnitude in corpus scope, and the corpus/video gap identifies cross-video confusion, not within-video imprecision, as the dominant error mode.

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RQ2. Five ablation results answer the embedding-strategy question. (i) Naive late fusion cannot rescue a weak text channel at any weight. (ii) Query decomposition buys recall at depth while visual re-ranking buys top-rank precision, and the two compose. (iii) SigLIP is the strongest index backbone, nearly doubling baseline R@1. (iv) A two-stage cheap-index/expensive-re-rank design matches the expensive index outright. (v) The modality ordering itself is domain-conditional: on YouCook2 (§5.5) transcript retrieval doubles visual retrieval and late fusion becomes the best configuration, exactly as the causal analysis predicted.

RQ3. A LangGraph agent with temporal tools and self-reflection improves QA accuracy by 10 points over a single-loop agent (McNemar $p = .031$), and a controlled evidence-channel experiment shows visual evidence nearly doubling temporal-next accuracy ($p = .004$). Modality therefore matters at the answering stage independently of retrieval.

Calibration. Prior-only baselines reframe every QA number above. Both models score .560 with no evidence at all. The best text-only stack merely matches that prior, and on temporal-next questions falls .227 below it ($p = .031$), while the multimodal stack exceeds its own prior by +.167 ($p = .0005$). Visual evidence is the only channel measured that beats the bare model.

Combining the best-measured component in every slot yields .727 five-way accuracy — at, and in point estimate above, the published zero-shot state of the art on this sample — while temporal grounding proves competitive with published weakly-supervised answer-grounding methods under a stricter, citation-producing protocol. Chapter 6 discusses the limitations and generality of these findings.

6 Discussion

6.1 Synthesis: three separable problems

The clearest lesson of Chapter 5 is that the end-to-end failures of a video RAG system decompose into three problems that are *separable* — they have different causes, respond to different interventions, and the interventions compose.

The first is a **recall problem**: the correct moment never enters the candidate pool because the query is phrased in causal or temporal language that a contrastive vision-language encoder cannot match against pixels. Query decomposition addresses exactly this — rewriting the question into scene captions lifted deep-rank recall (corpus R@10 +10% relative) while leaving the top rank untouched (§5.3.2). The second is a **precision problem**: the correct moment is in the pool but near-tied cosine scores cannot order it first. Second-stage re-scoring with a stronger visual model addresses this, and only this (§5.3.3); so does exchanging the index backbone outright (§5.3.4). The third is an **evidence problem**: the correct moment is retrieved, correctly time-stamped, and placed in front of the answerer — which then cannot answer because the decisive evidence is visual and its input is text. No amount of retrieval improvement fixes this; changing the evidence channel does (+87% on temporal-next questions, §5.4.2).

Reading the results this way explains why the interventions compose additively rather than cannibalising each other (Table 5.2): each removes a different bottleneck. It also offers a diagnostic recipe for practitioners: compare corpus against video scope to separate cross-video confusion from localisation error, compare shallow against deep recall to separate recall from precision problems, and compare text-only against multimodal answering to expose evidence starvation.

A second thread worth drawing out is that *the value of a modality is role- and question-dependent, not absolute*. The transcript channel failed twice as a *retrieval key*, once as embeddings (§5.2) and once as a cross-encoder signal (§5.3.3), and the prior-only

baselines of §5.4.3 discipline any remaining optimism about its third role as *evidence*: the best text-only stack (.547) merely matches the bare model’s answer prior (.560), and the apparent causal gains largely dissolve against that bar (CW .590 against a .574 prior).

What survives is sharply type-dependent. Transcript evidence genuinely beats the prior where speech describes the concurrent scene (TC .773 vs .545), and falls far below it where the answer is a silent visual action (TN .341 vs .568). The latter is an instructed-grounding effect, in which the model abandons a useful prior in favour of evidence that contains nothing. Speech in everyday video is a poor pointer to *where* something happens, and describes *what* happens only when someone happens to narrate it. Modality ablations that report a single “contribution” number therefore conflate roles and question types; the results above suggest both splits are necessary, and that any evidence-grounded system should be benchmarked against the bare model’s prior rather than against random.

Finally, the two-stage equivalence result (§5.3.4) has an architectural implication beyond this project. At 5,725 segments, indexing with the expensive model is trivially affordable, and yet the cheap-index/expensive-re-rank design already matched it to within measurement noise. Since re-ranking cost is fixed (top-30 per query) while indexing cost grows linearly with the corpus, the two-stage design is the only one of the two that survives scaling — and this work provides evidence that the quality sacrifice can be nil.

6.2 Limitations

Domain coverage. The primary conclusions are drawn from NExT-QA/GQA: short, everyday, speech-sparse home videos. The negative results about the text channel — fusion never helping (§5.3.1), the text cross-encoder hurting (§5.3.3) — are *conditional on this domain*, and that conditionality is now a measured result rather than a caveat: the speech-dense replication of §5.5 (YouCook2, 40 videos) reversed both signs exactly as predicted, with transcript retrieval doubling visual retrieval and late fusion becoming the best configuration. What remains limited is coverage: two domains, a small second corpus, and a retrieval-only replication — the positive results (decomposition, visual re-ranking, backbone ordering, evidence-channel effect) have not yet been re-tested outside NExT-QA, and the QA-stage experiments were not repeated on the second domain. Their transfer argument still rests on properties of contrastive encoders and of the QA task rather than on measurement.

A confound inside the text-channel result. The weak text channel has two entangled causes: the transcripts are often *uninformative* (chatter unrelated to the questions), and they are often *unreadable* by the predominantly-English CLIP text encoder (multilingual speech). The present experiments cannot apportion blame between the two. A translate-then-embed variant — machine-translating all transcripts to English before indexing — would separate them cheaply and was left undone only for time.

Segmentation granularity bounds temporal accuracy. Segments are fixed 8-second windows on a 4-second stride, while ground-truth intervals vary freely; even a perfect ranker cannot exceed the tIoU that the best-aligned window achieves. Part of the headline gap between $\tau = 0.5$ and $\tau = 0.3$ results (§5.3.5) is this quantisation, not ranking error. tIoU@1 therefore carries a structural ceiling that is a property of the segmenter; adaptive or shot-aligned segmentation would raise the ceiling itself.

Multiple-choice accuracy is a proxy. NExT-QA’s five-way format enables cheap, objective scoring, but it measures answer *selection*, not answer *quality*: the faithfulness of the generated prose and the correctness of its timestamp citations are audited only by the agent’s own reflection pass, not by human or LLM-as-judge evaluation. The white-dog case study (§5.4.2) illustrates that the generated answers can be precise and well-grounded, but no systematic open-ended-answer evaluation was performed.

Sample sizes and single runs. Retrieval results cover all 3,358 questions and are deterministic given the frozen decomposition cache. The QA experiments, however, are budget-limited: 150 questions for the agent comparison, 44 for the controlled vision experiment, and the text cross-encoder was rejected on a 100-question development subset. All agent numbers are single runs of non-deterministic API models. The two central paired comparisons are, however, statistically supported: exact McNemar tests on the shared question sets give $p = .031$ for the agent effect and $p = .004$ for the evidence-channel effect, with paired-bootstrap 95% CIs of $[+.013, +.187]$ and $[+.114, +.455]$ respectively (§5.4.1, §5.4.2) — so neither headline effect is plausibly run-to-run variance. The caution that remains is for the *small* differences in Table 5.5’s by-type columns ($n \leq 22$), which carry no such support and should not be over-read.

Asymmetries in the agent comparison. The controlled vision experiment (§5.4.2) ran under the *simple* agent, because at that point the LangGraph harness targeted OpenAI-compatible providers only; the graph-plus-vision combination was implemented subsequently and evaluated in §5.4.4, confirming the floor-not-ceiling reading on the

overall figure (.727 under the combined configuration). Its TN figure, however, is .568 — three questions *below* the simple agent’s .636 on the same items. That gap is within noise at $n = 44$, but it points at the asymmetry that remains: whether the reflection pass benefits from *seeing* the keyframes it audits, rather than the text rendering of the evidence it currently audits, has not been isolated, and TN is precisely the type where a text-mediated audit has least to offer.

No user study. The project plan scheduled a small user study (W10); it was descope in favour of the controlled vision experiment and the ablation matrix, as the more defensible evidence per unit of time available. The Streamlit demo (§3.6) stands in for qualitative validation, but claims about end-user utility remain unsupported and are deliberately absent from this dissertation.

6.3 Threats to validity

Construct validity. The hit criterion ($\text{correct video} \wedge \text{tIoU} \geq \tau$) is stricter than video-level retrieval metrics common in prior work; this deflates absolute numbers relative to that literature but is faithful to the moment-retrieval construct. For QA, the multiple-choice format admits answer-prior effects: a language model can reject implausible distractors without evidence. The controlled experiment bounds this — priors were identical across both evidence conditions, so the +29.5-point effect cannot be prior-driven — and the prior-only baselines (§5.4.3) measure the prior directly for both models (.560 each with no evidence at all), completing the decomposition: the .341 text-only TN figure turned out to sit *below* the model’s own prior, so what looked like weak evidence was evidence-induced suppression. This is why the dissertation rests its QA claims on paired within-model comparisons rather than absolute accuracies.

Internal validity. The decomposition cache is shared verbatim across all configurations that use it, eliminating decomposition variance from those comparisons. The re-ranker’s fusion weight (0.5) and candidate pool (30) were fixed a priori rather than tuned, and the development subset used to reject the text cross-encoder ($n = 100$) is disjoint in purpose but not in items from the full evaluation set; since the decision it informed was to *exclude* a component, any bias is conservative. Agent experiments called the same LLM family for decomposition and answering (DeepSeek), which could in principle correlate their errors; retrieval experiments are unaffected, as decomposition

quality is identical across all retrieval configurations.

External validity. Beyond the domain limitation above, two scale caveats apply. The corpus (567 videos, 5,725 segments) is small enough that exhaustive HNSW search is effectively exact; ANN recall degradation at millions of segments is not measured. And the two-stage equivalence claim is demonstrated at a scale where *both* designs are feasible — its extrapolation to large corpora rests on the argument of §6.1, not on direct measurement. Finally, all API-based results are tied to unversioned commercial model endpoints (deepseek-chat, claude-opus-4-8, mid-2026); the committed prompts, caches, and result files make the analysis auditable, but bit-exact replication of agent runs is not guaranteed by the providers.

6.4 Summary

The findings hold together as a single narrative: in everyday video, visual signal dominates at every stage where a modality choice exists — and the speech-dense replication shows this dominance is a measurable property of the domain, reversing cleanly where narration is dense and task-aligned. The remaining errors are separable into recall, precision, and evidence problems with independent, composable remedies, benchmarked against the bare model’s prior rather than against random. The principal limitations — partial domain coverage, proxy QA metrics, budget-limited agent samples, and the untested multimodal-reflection interaction — are all addressable with more compute rather than new machinery, and none threatens the direction of the reported effects, whose magnitudes (+69% relative R@1 from the backbone, +87% from the evidence channel, 2× modality reversal on YouCook2) far exceed the plausible noise floor. Chapter 7 concludes and sets out the corresponding future work.

7 Conclusion and Future Work

7.1 Answers to the research questions

RQ1 — retrieval effectiveness against text-only baselines. A multi-modal pipeline retrieving on visual embeddings outperforms transcript-based retrieval by an order of magnitude at moment granularity on everyday video (corpus-scope R@10 .111 vs .011 at $\text{tIoU} \geq 0.5$). The fully developed system, a SigLIP index with LLM query decomposition, nearly doubles that initial visual baseline in turn (R@1 .026 $\rightarrow$.048; video-scope $\text{tIoU}@1$.230). Scope analysis attributes the remaining difficulty to cross-video confusion rather than within-video localisation: given the right video, the system places a correct moment in its top five for two-thirds of questions at the looser $\tau = 0.3$ (R@5 .677). The answer to RQ1 is therefore affirmative with a precise caveat: *visual* retrieval is effective and improvable, while retrieval on what is *said* is, on this domain, not a viable baseline so much as a cautionary one. §5.5 shows that the caveat cuts both ways, since on speech-dense video the same pipeline’s transcript channel doubles its visual channel.

RQ2 — the multi-modal embedding strategy. The experiments support a strategy, not merely a ranking: index the visual channel with the strongest affordable contrastive backbone (SigLIP SO400M dominated both CLIP variants, +69% relative R@1 as the index); if the strongest backbone is unaffordable at corpus scale, index cheaply and re-rank its top candidates with the stronger model, which matched the expensive index to within measurement noise; translate questions into the encoder’s native register (scene captions) via LLM decomposition for deep-rank recall; and do *not* fuse in a weak text channel at any weight, nor re-rank with a text cross-encoder, since both degrade quality for an identified cause (sparse, multilingual, conversational speech). For the audio channel, which enters this system as transcribed speech (§1.2), the optimal strategy on the primary benchmark is the null strategy; on-screen-text OCR was out of scope and remains untested. The speech-dense replication (§5.5) then completes the

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answer. The ordering is domain-conditional by measurement rather than conjecture: on YouCook2 the transcript channel doubles the visual channel and late fusion becomes the best configuration. The optimal embedding strategy is therefore not a fixed recipe but a measured, domain-adaptive choice, and this dissertation’s harness is the instrument that measures it.

RQ3 — agentic decomposition and synthesis. Query decomposition, temporal tool use, and bounded self-reflection each demonstrably contribute. At the retrieval layer, decomposition is safe-by-construction (the original query is always retained) and lifts recall. At the answering layer, a LangGraph state machine that anchors events, walks the timeline, and audits its own citations raises five-way QA accuracy from .447 to .547 over a single-loop agent, with the largest gains on causal and temporal-current questions. The prior-only baselines calibrate what that machinery buys: with text-only evidence even the better agent only recovers the bare model’s answer prior (.560), because the channel’s genuine gains (temporal-current) are cancelled by its losses on silent visual actions (temporal-next, where delivered transcripts drag the model below its own prior). The controlled evidence-channel experiment completes the answer: agentic machinery alone cannot compensate for evidence the model cannot perceive — supplying keyframes to the answerer nearly doubled temporal-next accuracy (.341 $\rightarrow$.636) with everything else held fixed, and the full multimodal stack is the only configuration to beat its model’s prior (+.167). An effective video-QA agent is therefore *jointly* a retrieval planner and a multimodal reader; either half alone leaves the other’s headroom unrealised.

7.2 Future work

Five directions follow directly from the limitations identified in Chapter 6, ordered roughly by evidence-per-effort.

Repair the text channel, then re-test fusion. Machine-translate transcripts to English (or adopt a multilingual text-retrieval encoder) before indexing, disentangling the informativeness/readability confound of §6.2 — and only then revisit late fusion and text re-ranking, whose negative results are conditional on the broken channel.

Scale the best-configuration evaluation. The graph-plus-multimodal combination now exists and reaches .727 on the 150-question sample (§5.4.4); the direct next step is statistical rather than architectural — evaluating the full grounded split (approximately

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US\$250) would shrink the confidence interval from ± 7 to ± 1.5 points and settle the comparison with published zero-shot systems, and a multimodal *reflection* pass (auditing the keyframes themselves) is the one untested component interaction.

Complete the open-source model comparison. The provider abstraction already targets any OpenAI-compatible endpoint; running the QA suite against local Qwen/Llama models (the plumbing and evaluation scripts exist) would quantify the API-versus-open trade-off the project plan anticipated.

Adaptive segmentation, on a tested segmenter. Fixed 8-second windows impose a structural ceiling on tIoU (§6.2); shot-aligned or query-adaptive windowing would raise the ceiling itself, and the evaluation harness can measure the gain directly. Reworking the segmenter is also the point at which the missing unit tests (§4.7) should be written: the boundary arithmetic and the rank-fusion and imputation logic are exactly the code whose errors surface as plausible-looking metrics rather than crashes.

Deepen the second domain, and run the user study. The speech-dense retrieval replication is done (§5.5); its natural extensions are to repeat the QA-stage and agent experiments on YouCook2, replicate the *positive* results (decomposition, re-ranking, backbone ordering) cross-domain, and derive a quantitative domain-adaptive fusion policy from measured speech density. The descoped user study remains the right instrument for claims about end-user utility that no offline metric can support.

7.3 Closing remark

The project set out to make a moment of video as findable as a sentence of text, under the discipline that every design choice be measured. Its most useful legacy may be less the headline numbers than the shape of the evidence: failures separated into recall, precision, and evidence problems; negative results retained with their causes; and a pipeline in which each remedy could be switched on, switched off, and priced. Video understanding systems will continue to change quickly; an evaluation harness that can say *which part helped, by how much, and why* will remain the transferable asset.

A Full per-question-type result tables

Generated by `make_appendix_b.py` from the committed result files in `results/`; regenerate after any re-run. Question types: CW/CH causal, TN/TC/TP temporal. TP has only 52 grounded instances corpus-wide (2 in the 150-question QA sample) — its rows are reported for completeness and should not be interpreted.

A.1 ViT-B-32 baseline — corpus scope (visual, tau 0.5)

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.026	.075	.111	.047	.117	.035
CW	1474	.022	.063	.097	.041	.103	.030
(causal why)							
TN	879	.025	.073	.107	.045	.107	.032
(tempo- ral next)							
CH	481	.037	.116	.156	.070	.175	.051
(causal how)							
TC	472	.030	.078	.121	.052	.124	.041
(tempo- ral current)							

A Full per-question-type result tables

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
TP (tempo- ral previ- ous)	52	.000	.019	.096	.013	.070	.015

A.2 ViT-B-32 baseline — video scope (visual, tau 0.5)

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.148	.394	.491	.250	.659	.203
CW (causal why)	1474	.149	.394	.491	.250	.655	.202
TN (tempo- ral next)	879	.129	.379	.495	.230	.621	.177
CH (causal how)	481	.156	.403	.491	.258	.710	.224
TC (tempo- ral current)	472	.182	.424	.494	.283	.697	.235
TP (tempo- ral previ- ous)	52	.096	.308	.442	.187	.585	.151

A.3 Decomposition + visual re-ranking — corpus scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.030	.087	.128	.055	.135	.040
CW	1474	.028	.077	.121	.051	.125	.036
(causal why)							
TN	879	.031	.078	.107	.051	.119	.034
(tempo- ral next)							
CH	481	.037	.129	.170	.073	.188	.061
(causal how)							
TC	472	.030	.097	.144	.061	.143	.044
(tempo- ral current)							
TP	52	.019	.058	.115	.035	.110	.031
(tempo- ral previ- ous)							

A.4 Decomposition + visual re-ranking — video scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.154	.410	.501	.258	.670	.209
CW	1474	.152	.410	.496	.259	.667	.206
(causal why)							

A Full per-question-type result tables

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
TN (tempo- ral next)	879	.133	.396	.513	.238	.631	.183
CH (causal how)	481	.158	.414	.497	.264	.719	.235
TC (tempo- ral current)	472	.195	.439	.508	.293	.704	.243
TP (tempo- ral previ- ous)	52	.135	.346	.442	.212	.627	.184

A.5 SigLIP + decomposition (final best) — corpus scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.048	.110	.161	.076	.167	.055
CW (causal why)	1474	.054	.115	.157	.080	.161	.058
TN (tempo- ral next)	879	.027	.081	.140	.054	.152	.039
CH (causal how)	481	.079	.156	.212	.114	.243	.092

A Full per-question-type result tables

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
TC (tempo- ral current)	472	.038	.110	.172	.072	.147	.043
TP (tempo- ral previ- ous)	52	.019	.019	.077	.027	.089	.028

A.6 SigLIP + decomposition (final best) — video scope

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
overall	3358	.176	.419	.496	.275	.686	.230
CW (causal why)	1474	.193	.427	.495	.289	.695	.243
TN (tempo- ral next)	879	.130	.412	.502	.243	.647	.195
CH (causal how)	481	.189	.410	.486	.280	.718	.247
TC (tempo- ral current)	472	.199	.424	.504	.293	.704	.242

A Full per-question-type result tables

Type	n	R@1	R@5	R@10	MRR	nDCG@10	tIoU@1
TP (tempo- ral previ- ous)	52	.135	.365	.423	.207	.655	.194

A.7 QA accuracy by type — simple agent, text-only (150 questions)

Type	n	Accuracy
overall	150	.447
CW (causal why)	61	.459
TN (temporal next)	44	.341
CH (causal how)	21	.429
TC (temporal current)	22	.591
TP (temporal previous)	2	1.000

A.8 QA accuracy by type — graph agent, text-only (150 questions)

Type	n	Accuracy
overall	150	.547
CW (causal why)	61	.590
TN (temporal next)	44	.341
CH (causal how)	21	.619
TC (temporal current)	22	.773
TP (temporal previous)	2	.500

A.9 QA accuracy — multimodal simple agent (44 TN questions)

Type	n	Accuracy
overall	44	.636
TN (temporal next)	44	.636

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